

Tony Tae-Hyoung KIM

associate professor

基本信息

姓名: Tony Tae-Hyoung KIM 所属院校: 南洋理工大学
地区: Singapore 海外经历: 有

学科领域

集成电路设计与集成系统 微电子科学与工程 电子与计算机工程

研究领域

低功耗电路与能量管理 存储器设计与架构 数字集成电路设计 嵌入式系统与硬件设计
低功耗设计与能耗优化 模拟与数模混合电路 计算机体系结构 超大规模集成电路设计
微电子器件与制造

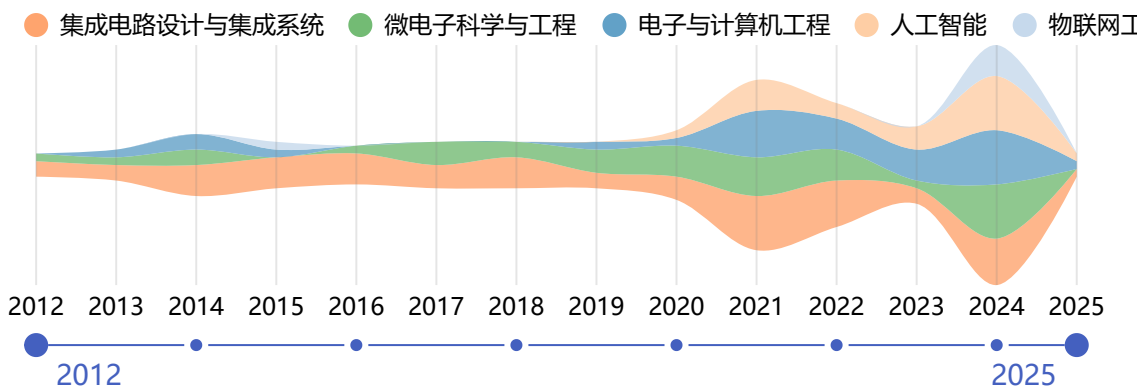
个人简介

Tony Tae-Hyoung KIM is an Associate Professor at Nanyang Technological University, Singapore. His research focuses on ultra-low power IC design, computing-in-memory, and tiny machine learning accelerators. With a strong publication record and significant citations, his work has contributed significantly to the field of high-density cache memory, subthreshold SRAM, and energy-efficient circuit design. His research also encompasses reliability monitoring, voltage scalability, and energy harvesting systems, making substantial impacts in both academia and industry.

工作经历

Nanyang Technological University - Associate Professor

近年发表的论文



A Graph-Based Accelerator of Retinex Model With Bit-Serial Computing for Image Enhancements

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Wei, Zhengzhe; Mu, Junjie; Zheng, Yuanjin; Kim, Tony Tae-Hyoung; Kim, Bongjin

被引用次数: 0 2025

A 65-nm 55.8-TOPS/W Compact 2T eDRAM-Based Compute-in-Memory Macro With Linear Calibration

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Zhang, Xueyong; Jo, Yong-Jun; Kim, Tony Tae-Hyoung

被引用次数: 0 2025

An Area-Efficient Dual-Mode SAR ADC With 3-Level Analog Prediction to Compensate for Input Range Loss of Mismatch Error Shaping

2024 50TH IEEE EUROPEAN SOLID-STATE ELECTRONICS RESEARCH CONFERENCE, ESSERC 2024

Seong, Kiho; Seo, Jee-Taeck; Lee, Jeonghun; Lee, Sukho; Kim, Tony Tae-Hyoung; Baek, Kwang-Hyun

被引用次数: 0 2024

A Dual 7T SRAM-Based Zero-Skipping Compute-In-Memory Macro With 1-6b Binary Searching ADCs for Processing Quantized Neural Networks

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Yu, Chengshuo; Jiang, Haoge; Mu, Junjie; Chai, Kevin Tshun Chuan; Kim, Tony Tae-Hyoung; Kim, Bongjin

被引用次数: 0 2024

A 2.793 μ W Near-Threshold Neuronal Population Dynamics Trajectory Filter for Reliable Simultaneous Localization and Mapping

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Wei, Zhengzhe; Dong, Boyi; Su, Yuqi; Wang, Yi; Yang, Chuanshi; Lu, Yuncheng; Wang, Chao; Kim, Tony Tae-Hyoung; Zheng, Yuanjin

被引用次数: 0 2024

A 4.2pJ/Pixel 480 fps Stereo Vision Processor with Pixel Level Pipelined Architecture and Two-path Aggregation Semi-Global Matching

2024 IEEE CUSTOM INTEGRATED CIRCUITS CONFERENCE, CICC

Li, Zehao; Lu, Yuncheng; Anh Tuan Do; Kim, Tony Tae-Hyoung

被引用次数: 0

2024

Live Demonstration: Real-Time Object Detection & Classification System in IoT with Dynamic Neuromorphic Vision Sensors

2024 IEEE INTERNATIONAL SYMPOSIUM ON CIRCUITS AND SYSTEMS, ISCAS 2024

Li, Zehao; Lu, Wenhao; Lu, Yuncheng; Li, Junying; Shi, Yucen; Zheng, Yuanjin; Kim, Tony Tae-Hyoung

被引用次数: 0

2024

An Energy-Efficient Object Detection System in IoT with Dynamic Neuromorphic Vision Sensors

2024 IEEE INTERNATIONAL SYMPOSIUM ON CIRCUITS AND SYSTEMS, ISCAS 2024

Li, Zehao; Lu, Wenhao; Lu, Yuncheng; Li, Junying; Shi, Yucen; Zheng, Yuanjin; Kim, Tony Tae-Hyoung

被引用次数: 0

2024

A Memory -Efficient High -Speed Event -based Object Tracking System

2024 IEEE INTERNATIONAL SYMPOSIUM ON CIRCUITS AND SYSTEMS, ISCAS 2024

Lu, Yuncheng; Cui, Kaixiang; Shi, Yucen; Li, Zehao; Li, Junying; Lu, Wenhao; Zheng, Yuanjin; Kim, Tony Tae-Hyoung

被引用次数: 0

2024

A 2.793 μ W Near-Threshold Neuronal Population Dynamics Simulator for Reliable Simultaneous Localization and Mapping

2024 IEEE INTERNATIONAL SYMPOSIUM ON CIRCUITS AND SYSTEMS, ISCAS 2024

Wei, Zhengzhe; Dong, Boyi; Su, Yuqi; Wang, Yi; Yang, Chuanshi; Lu, Yuncheng; Wang, Chao; Kim, Tony Tae-Hyoung; Zheng, Yuanjin

被引用次数: 1

2024

A 0.6-to-1.2 V Scaling-Friendly Discrete-Time OTA-Free Linear VCO-Based TEXPRESERVE0 ADC Suitable for DVFS

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

An, Kyung-Chan; Narasimman, Neelakantan; Kim, Tony Tae-Hyoung

被引用次数: 1

2024

Reference-Free Dual-Mode Cell-to-Cell Sensing for Area-Efficient Resistive Memory

2024 IEEE THE 20TH ASIA PACIFIC CONFERENCE ON CIRCUITS AND SYSTEMS, APCCAS 2024

An, Byung-Kwon; Kim, Tony Tae-Hyoung

被引用次数: 0

2024

A 2.5-GHz Dynamic Performance-Enhanced Nonlinear DAC-Based Direct-Digital Frequency Synthesizer in 65-nm CMOS Process

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Yoon, Dong-Hyun; Baek, Kwang-Hyun; Kim, Tony Tae-Hyoung

被引用次数: 1

2024

FlexSpin: A CMOS Ising Machine With 256 Flexible Spin Processing Elements With 8-b Coefficients for Solving Combinatorial Optimization Problems

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Su, Yuqi; Kim, Tony Tae-Hyoung; Kim, Bongjin

被引用次数: 3

2024

HGRP: A 181- μ W Real-Time Hand Gesture Recognition Processor Based on Bi-Directional Convolution and Iteration-Free Feature Clustering

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Lu, Yuncheng; Li, Zehao; Kim, Tony Tae-Hyoung

被引用次数: 0

2024

A Scalable and Reconfigurable Bit-Serial Compute-Near-Memory Hardware Accelerator for Solving 2-D/3-D Partial Differential Equations

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Mu, Junjie; Yu, Chengshuo; Kim, Tony Tae-Hyoung; Kim, Bongjin

被引用次数: 1

2024

SESOMP: A Scalable and Energy-Efficient Self-Organizing Map Processor for IoT Devices

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Lu, Yuncheng; Zhang, Xin; Wang, Bo; Kim, Tony Tae-Hyoung

被引用次数: 0

2024

A High-Resolution Discrete-Time Second-Order $\Sigma\Delta$ ADC with Improved Tolerance to KT/C Noise Using Low Oversampling Ratio

SENSORS

An, Kyung-Chan; Narasimman, Neelakantan; Kim, Tony Tae-Hyoung

被引用次数: 0

2024

Design of a Reliable Current Sense Amplifier with Dynamic Reference for Resistive Memory

JOURNAL OF SEMICONDUCTOR TECHNOLOGY AND SCIENCE

An, Byung-Kwon; Zhang, Xueyong; Do, Anh Tuan; Kim, Tony Tae-Hyoung

被引用次数: 0

2024

A Reconfigurable Step-Down Switched-Capacitor Power Converter Using Optimized Partial Series-Parallel (OPSP) Topology

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Jang, Chunghee; Shin, Dong-Hyun; Kim, Young-Kyu; Kim, Tony Tae-Hyoung; Lee, Sukho; Baek, Kwang-Hyun

被引用次数: 0

2024

A 1-Mb RRAM Macro With 9.8 ns Read Access Time Utilizing Dynamic Reference Voltage for Reliable Sensing Operation

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Mu, Junjie; Lu, Lu; Kim, Ju Eon; An, Byungkwon; Sharma, Vishal; Lekshmi, Arya Jagath; Dananjaya, Putu Andhita; Lai, Weng Hong; Lew, Wen Siang; Kim, Tony Tae-Hyoung

被引用次数: 0

2024

A Robust Time-Based Multi-Level Sensing Circuit for Resistive Memory

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Zhang, Xueyong; An, Byung-Kwon; Kim, Tony Tae-Hyoung

被引用次数: 4

2023

A 1-16b Reconfigurable 80Kb 7T SRAM-Based Digital Near-Memory Computing Macro for Processing Neural Networks

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Kim, Hyunjoon; Mu, Junjie; Yu, Chengshuo; Kim, Tony Tae-Hyoung; Kim, Bongjin

被引用次数: 12

2023

BP-SCIM: A Reconfigurable 8T SRAM Macro for Bit-Parallel Searching and Computing In-Memory

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Chen, Yuzong; Mu, Junjie; Kim, Hyunjoon; Lu, Lu; Kim, Tony Tae-Hyoung

被引用次数: 10

2023

A Digital Bit-Reconfigurable Versatile Compute-In-Memory Macro for Machine Learning Acceleration

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Zhang, Xin; Lu, Yuncheng; Wang, Bo; Kim, Tony Tae-Hyoung

被引用次数: 1

2023

A 184- μ W Error-Tolerant Real-Time Hand Gesture Recognition System With Hybrid Tiny Classifiers Utilizing Edge CNN

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Lu, Yuncheng; Le, Van Loi; Kim, Tony Tae-Hyoung

被引用次数: 6

2023

A Time-Domain Wavefront Computing Accelerator With a 32 x 32 Reconfigurable PE Array

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Yu, Chengshuo; Mu, Junjie; Su, Yuqi; Chai, Kevin Tshun Chuan; Kim, Tony Tae-Hyoung; Kim, Bongjin

被引用次数: 1

2023

A Reconfigurable 8T SRAM Macro for Bit-Parallel Searching and Computing In-Memory

2022 IEEE INTERNATIONAL SYMPOSIUM ON CIRCUITS AND SYSTEMS (ISCAS 22)

Chen, Yuzong; Mu, Junjie; Kim, Hyunjoon; Lu, Lu; Kim, Tony Tae-Hyoung

被引用次数: 2

2022

Guest Editorial Revolution of AI and Machine Learning With Processing-in-Memory (PIM): From Systems, Architectures, to Circuits

IEEE JOURNAL ON EMERGING AND SELECTED TOPICS IN CIRCUITS AND SYSTEMS

Kim, Tony Tae-Hyoung; Kim, Bongjin; Kim, Joo-Young; Kulkarni, Jaydeep P.

An Overview of Processing-in-Memory Circuits for Artificial Intelligence and Machine Learning

IEEE JOURNAL ON EMERGING AND SELECTED TOPICS IN CIRCUITS AND SYSTEMS

Kim, Donghyuk; Yu, Chengshuo; Xie, Shanshan; Chen, Yuzong; Kim, Joo-Young; Kim, Bongjin; Kulkarni, Jaydeep P.; Kim, Tony Tae-Hyoung

被引用次数: 29

2022

A Reconfigurable 16Kb AND8T SRAM Macro With Improved Linearity for Multibit Compute-In Memory of Artificial Intelligence Edge Devices

IEEE JOURNAL ON EMERGING AND SELECTED TOPICS IN CIRCUITS AND SYSTEMS

Sharma, Vishal; Kim, Ju-Eon; Kim, Hyunjoon; Lu, Lu; Kim, Tony Tae-Hyoung

被引用次数: 17

2022

A 3.2-GHz 178-fs σ_{rms} Jitter Subsampling PLL/DLL-Based Injection-Locked Clock Multiplier

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Yoon, Dong-Hyun; Jung, Dong-Kyu; Seong, Kiho; Han, Jae-Soub; Chung, Keun-Yong; Kim, Ju-Eon; Kim, Tony Tae-Hyoung; Baek, Kwang-Hyun

被引用次数: 1

2022

A High Reliable SRAM-Based PUF With Enhanced Challenge-Response Space

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Lu, Lu; Kim, Tony Tae-Hyoung

被引用次数: 10

2022

A Robust Time-Based Multi-Level Sensing Circuit for Resistive Memory

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Zhang, Xueyong; An, Byung-Kwon; Kim, Tony Tae-Hyoung

被引用次数: 0

2022

A 65-nm 8T SRAM Compute-in-Memory Macro With Column ADCs for Processing Neural Networks

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Yu, Chengshuo; Yoo, Taegeun; Chai, Kevin Tshun Chuan; Kim, Tony Tae-Hyoung; Kim, Bongjin

被引用次数: 40

2022

A 6T SRAM Based Two-Dimensional Configurable Challenge-Response PUF for Portable Devices

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Lu, Lu; Yoo, Taegeun; Kim, Tony Tae-Hyoung

被引用次数: 7

2022

A 64 Kb Reconfigurable Full-Precision Digital ReRAM-Based Compute-In-Memory for Artificial Intelligence Applications

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Sharma, Vishal; Kim, Hyunjoon; Kim, Tony Tae-Hyoung

被引用次数: 8

2022

A Low-Power Gesture Recognition System utilizing Hybrid Tiny Classifiers

2022 19TH INTERNATIONAL SOC DESIGN CONFERENCE (ISOCC)

Lu, Yuncheng; Li, Zehao; Zhang, Xin; Kim, Tony Tae-Hyoung

被引用次数: 0

2022

A Capacitance to Digital Converter Using Continuous Time $\Delta\Sigma$ Modulator for Microphone-Based Auscultation

IEEE SENSORS JOURNAL

Narasimman, Neelakantan; Nag, Dipankar; Chai, Kevin T. C.; Kim, Tony Tae-Hyoung

被引用次数: 10

2021

A 0.6-to-1.2 V Scaling Friendly Discrete-Time OTA-Free $\Delta\Sigma$ -ADC for IoT Applications

ESSCIRC 2021 - IEEE 47TH EUROPEAN SOLID STATE CIRCUITS CONFERENCE (ESSCIRC)

An, Kyung-Chan; Narasimman, Neelakantan; Kim, Tony Tae-Hyoung

被引用次数: 2

2021

Guest Editorial Special Issue on Selected Papers From ISCAS 2021

IEEE TRANSACTIONS ON BIOMEDICAL CIRCUITS AND SYSTEMS

John, Deepu; Tae-Hyoung Kim, Tony; Johnston, Matthew L.

被引用次数: 0

2021

A Programmable 6T SRAM-Based PUF with Dynamic Stability Data Masking

IEEE ASIAN SOLID-STATE CIRCUITS CONFERENCE (A-SSCC 2021)

Lu, Lu; Kim, Tony Tae-Hyoung

被引用次数: 0

2021

A 3.2 GHz 178fsrms Jitter Injection Locked Clock Multiplier Using Sub-Sampling FTL and DLL for In- Band Noise Improvement

IEEE ASIAN SOLID-STATE CIRCUITS CONFERENCE (A-SSCC 2021)

Yoon, Dong-Hyun; Jung, Dong-Kyu; Seong, Kiho; Eom, Tae-Hyeok; Han, Jae-Soub; Kim, Ju Eon; Kim, Tony Tae-Hyoung; Baek, Kwang-Hyun

被引用次数: 1

2021

A Zero-Skipping Reconfigurable SRAM In-Memory Computing Macro with Binary-Searching ADC

IEEE 51ST EUROPEAN SOLID-STATE DEVICE RESEARCH CONFERENCE (ESSDERC 2021)

Yu, Chengshuo; Chai, Kevin Tshun Chuan; Kim, Tony Tae-Hyoung; Kim, Bongjin

被引用次数: 12

2021

An Ultra-Low-power Real-Time Hand-Gesture Recognition System for Edge Applications

2021 IEEE 3RD INTERNATIONAL CONFERENCE ON ARTIFICIAL INTELLIGENCE CIRCUITS AND SYSTEMS (AICAS)

Lu, Yuncheng; Li, Zehao; Kim, Tony Tae-Hyoung

被引用次数: 1

2021

An Ultra-Low-Voltage VCO-Based $\Delta\Sigma$ Modulator Using Self-compensated Current Reference for Variation Tolerance

CIRCUITS SYSTEMS AND SIGNAL PROCESSING

Narasimman, Neelakantan; Kim, Tony Tae-Hyoung

被引用次数: 1

2021

A 16-kb 9T Ultralow-Voltage SRAM With Column-Based Split Cell-VSS, Data-Aware Write-Assist, and Enhanced Read Sensing Margin in 28-nm FDSOI

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Siddiqui, M. Sultan M.; Lee, Zhao Chuan; Kim, Tony Tae-Hyoung

被引用次数: 10

2021

Zero-Skipping Reconfigurable SRAM In-Memory Computing Macro with Binary-Searching ADC

ESSCIRC 2021 - IEEE 47TH EUROPEAN SOLID STATE CIRCUITS CONFERENCE (ESSCIRC)

Yu, Chengshuo; Chai, Kevin Tshun Chuan; Kim, Tony Tae-Hyoung; Kim, Bongjin

被引用次数: 0

2021

Tear-Based Aqueous Batteries for Smart Contact Lenses Enabled by Prussian Blue Analogue Nanocomposites

NANO LETTERS

Yun, Jeonghun; Zeng, Yongpeng; Kim, Moobum; Gao, Caitian; Kim, Yeongae; Lu, Lu; Kim, Tony Tae-Hyoung; Zhao, Wenting; Bae, Tae-Hyun; Lee, Seok Woo

被引用次数: 27

2021

A 0.007 mm² 0.6 V 6 MS/s Low-Power Double Rail-to-Rail SAR ADC in 65-nm CMOS

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Jo, Yong-Jun; Kim, Ju Eon; Baek, Kwang-Hyun; Kim, Tony Tae-Hyoung

被引用次数: 4

2021

Colonnade: A Reconfigurable SRAM-Based Digital Bit-Serial Compute-In-Memory Macro for Processing Neural Networks

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Kim, Hyunjoon; Yoo, Taegeun; Kim, Tony Tae-Hyoung; Kim, Bongjin

被引用次数: 84

2021

A Logic-Compatible eDRAM Compute-In-Memory With Embedded ADCs for Processing Neural Networks

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Yu, Chengshuo; Yoo, Taegeun; Kim, Hyunjoon; Kim, Tony Tae-Hyoung; Chuan, Kevin Chai Tshun; Kim, Bongjin

被引用次数: 48

2021

A Power-Aware Toggling-Frequency Actuator in Data-Toggling SRAM for Secure Data Protection

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Ho, Weng-Geng; Chong, Kwen-Siong; Kim, Tony Tae-Hyoung; Gwee, Bah-Hwee

被引用次数: 0

2021

SRAM Radiation Hardening Through Self-Refresh Operation and Error Correction

IEEE TRANSACTIONS ON DEVICE AND MATERIALS RELIABILITY

Siddiqui, M. Sultan M.; Ruchi, Sharma; Loi Van Le; Yoo, Taegeun; Chang, Ik-Joon; Kim, Tony Tae-Hyoung

被引用次数: 9

2020

A triboelectric nanogenerator energy harvesting system based on load-aware control for input power from 2.4 μ W to 15.6 μ W

NANO ENERGY

Rawy, Karim; Sharma, Ruchi; Yoon, Hong-Joon; Khan, Usman; Kim, Sang-Woo; Kim, Tony Tae-Hyoung

被引用次数: 13

2020

Energy-Efficient Data-Aware SRAM Design Utilizing Column-Based Data Encoding

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Do, Anh Tuan; Zeinolabedin, Seyed Mohammad Ali; Kim, Tony Tae-Hyoung

被引用次数: 11

2020

A 137- μ W 1.78-mm² 30-Frames/s Real-Time Gesture Recognition SoC for Smart Devices

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Le, Van Loi; Yoo, Taegeun; Kim, Ju Eon; Ba, Ngoc Le; Baek, Kwang-Hyun; Kim, Tony Tae-Hyoung

被引用次数: 3

2020

A 0.506-pJ 16-kb 8T SRAM With Vertical Read Wordlines and Selective Dual Split Power Lines

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Lu, Lu; Yoo, Taegeun; Le, Van Loi; Kim, Tony Tae-Hyoung

被引用次数: 9

2020

A 0.5 V 8-12 Bit 300 KSPS SAR ADC With Adaptive Conversion Time Detection-and-Control for High Immunity to PVT Variations

IEEE ACCESS

Kim, Ju Eon; Yoo, Taegeun; Jung, Dong-Kyu; Yoon, Dong-Hyun; Seong, Kiho; Kim, Tony Tae-Hyoung; Baek, Kwang-Hyun

被引用次数: 6

2020

Time-Interleaved SAR ADC with Background Timing-Skew Calibration for UWB Wireless Communication in IoT Systems

SENSORS

Seong, Kiho; Jung, Dong-Kyu; Yoon, Dong-Hyun; Han, Jae-Soub; Kim, Ju-Eon; Kim, Tony Tae-Hyoung; Lee, Woojoo; Baek, Kwang-Hyun

被引用次数: 14

2020

An Area-Efficient 128-Channel Spike Sorting Processor for Real-Time Neural Recording With 0.175 μ W/Channel in 65-nm CMOS

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Anh Tuan Do; Zeinolabedin, Seyed Mohammad Ali; Jeon, Dongsuk; Sylvester, Dennis; Kim, Tony Tae-Hyoung

被引用次数: 33

2019

A 256 pixel, 21.6 μ W infrared gesture recognition processor for smart devices

MICROELECTRONICS JOURNAL

Le Ba, Ngoc; Oh, Sechang; Sylvester, Dennis; Kim, Tony Tae-Hyoung

被引用次数: 5

2019

Improving uniformity and reliability of SRAM PUFs utilizing device aging phenomenon for unique identifier generation

MICROELECTRONICS JOURNAL

Garg, Achiranshu; Lee, Zhao Chuan; Lu, Lu; Kim, Tony Tae-Hyoung

被引用次数: 10

2019

An 8T SRAM With On-Chip Dynamic Reliability Management and Two-Phase Write Operation in 28-nm FDSOI

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Lee, Zhao Chuan; Siddiqui, M. Sultan M.; Kong, Zhi-Hui; Kim, Tony Tae-Hyoung

被引用次数: 2

2019

A Secure Data-Toggling SRAM for Confidential Data Protection

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Ho, Weng-Geng; Chong, Kwen-Siong; Kim, Tony Tae-Hyoung; Gwee, Bah-Hwee

被引用次数: 11

2019

An Area Efficient 1024-Point Low Power Radix-22 FFT Processor With Feed-Forward Multiple Delay Commutators

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Ba, Ngoc Le; Kim, Tony Tae-Hyoung

被引用次数: 19

2018

A 12-bit Multi-Channel R-R DAC Using a Shared Resistor String Scheme for Area-Efficient Display Source Driver

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Jung, Dong-Kyu; Jung, Yun-Hwan; Yoo, Taegeun; Yoon, Dong-Hyun; Jung, Bo-Yun; Kim, Tony Tae-Hyoung; Baek, Kwang-Hyun

被引用次数: 14

2018

A 32 kb 9T near-threshold SRAM with enhanced read ability at ultra-low voltage operation

SOLID-STATE ELECTRONICS

Kim, Tony Tae-Hyoung; Lee, Zhao Chuan; Anh Tuan Do

被引用次数: 5

2018

A 0.016 mV/mA Cross-Regulation 5-Output SIMO DC-DC Buck Converter Using Output-Voltage-Aware Charge Control Scheme

IEEE TRANSACTIONS ON POWER ELECTRONICS

Pham, Ngoc-Son; Yoo, Taegeun; Kim, Tony Tae-Hyoung; Lee, Chan-Gun; Baek, Kwang-Hyun

被引用次数: 37

2018

An 88% Efficiency 0.1-300- μ W Energy Harvesting System With 3-D MPPT Using Switch Width Modulation for IoT Smart Nodes

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Rawy, Karim; Yoo, Taegeun; Kim, Tony Tae-Hyoung

被引用次数: 59

2018

A 0.4-V, 0.138-fJ/Cycle Single-Phase-Clocking Redundant-Transition-Free 24T Flip-Flop With Change-Sensing Scheme in 40-nm CMOS

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Le, Van Loi; Li, Juhui; Chang, Alan; Kim, Tony Tae-Hyoung

被引用次数: 16

2018

An Area and Energy Efficient Ultra-Low Voltage Level Shifter With Pass Transistor and Reduced-Swing Output Buffer in 65-nm CMOS

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Van Loi Le; Kim, Tony Tae-Hyoung

被引用次数: 47

2018

A 0.4 V 12T 2RW dual-port SRAM with suppressed common-row-access disturbance

MICROELECTRONICS JOURNAL

Wang, Bo; Zhou, Jun; Kim, Tony Tae-Hyoung

被引用次数: 1

2017

A 220-mV Power-on-Reset Based Self-Starter With 2-nW Quiescent Power for Thermoelectric Energy Harvesting Systems

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Das, Abhik; Gao, Yuan; Kim, Tony Tae-Hyoung

被引用次数: 32

2017

A Self-Adaptive Time-Based MPPT With 96.2% Tracking Efficiency and a Wide Tracking Range of 10 μ A to 1 mA for IoT Applications

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Rawy, Karim; Kalathiparambil, Felix; Maurath, Dominic; Kim, Tony Tae-Hyoung

被引用次数: 30

2017

Yield Enhancement of Face-to-Face Cu-Cu Bonding With Dual-Mode Transceivers in 3DICs

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Aung, Myat-Thu-Linn; Yoshikawa, Takefumi; Tan, Chuan-Seng; Kim, Tony Tae-Hyoung

被引用次数: 5

2017

Design of Temperature-Aware Low-Voltage 8T SRAM in SOI Technology for High-Temperature Operation (25 $^{\circ}$ C-300 $^{\circ}$ C)

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Ngoc Le Ba; Kim, Tony Tae-Hyoung

被引用次数: 2

2017

A Power and Area Efficient Ultra-Low Voltage Laplacian Pyramid Processing Engine With Adaptive Data Compression

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Zeinolabedin, Seyed Mohammad Ali; Zhou, Jun; Kim, Tony Tae-Hyoung

被引用次数: 0

2016

Read Bitline Sensing and Fast Local Write-Back Techniques in Hierarchical Bitline Architecture for Ultralow-Voltage SRAMs

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Wang, Bo; Li, Qi; Kim, Tony Tae-Hyoung

被引用次数: 2

2016

2.31-Gb/s/ch Area-Efficient Crosstalk Canceled Hybrid Capacitive Coupling Interconnect for 3-D Integration

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Myat-Thu-Linn Aung; Lim, Teck Heng; Yoshikawa, Takefumi; Kim, Tony Tae-Hyoung

被引用次数: 13

2016

0.2 V 8T SRAM With PVT-Aware Bitline Sensing and Column-Based Data Randomization

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Do, Anh Tuan; Lee, Zhao Chuan; Wang, Bo; Chang, Ik-Joon; Liu, Xin; Kim, Tony Tae-Hyoung

被引用次数: 20

2016

SRAM devices and circuits optimization toward energy efficiency in multi-V_{th} CMOS

MICROELECTRONICS JOURNAL

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